

COMPLETED WORK

Firmware Development – Individual Sensors

- Developed and validated standalone firmware for MAX30101 + MAX32664 (heart rate, SpO₂, PPG)
- Implemented TMP117 temperature sensor driver with readings
- Integrated BMI270 IMU with accelerometer and gyroscope functionality
- Established I2C communication bus management for multi-sensor operation

System Integration – Foundation

- Created unified firmware architecture reading all three sensors simultaneously
- Verified sensor address allocation and bus arbitration (0x48, 0x57, 0x68)
- Validated data output via SWD debugging and serial terminal

REMAINING WORK

1. Verify sensor readings:

- Verify that sensor readings are accurate
- Perform necessary calibration if required

2. Bluetooth Low Energy Integration

- Implement BLE 5.0 stack using Nordic nRF Connect SDK
- Develop custom GATT profile for health data services (Heart Rate, Temperature, Motion)
- Configure BLE notifications and characteristics for real-time data streaming
- Implement pairing, bonding, and security protocols

3. Power Optimization & Battery Life Validation

- Implement adaptive sensor sampling algorithms
- Optimize BLE connection intervals for minimal power consumption
- Develop deep sleep mode with wake-on-motion capability
- Validate 2-3+ day battery life target with LiPo cell
- Monitor battery level and report via BLE.

4. Gesture Recognition & Haptic Feedback

- Develop tap and double-tap detection algorithms using BMI270 data
- Implement wrist-wear detection logic (motion + temperature fusion)
- Integrate vibration motor control for haptic alerts and confirmations

5. Over-the-Air Firmware Updates

- Integrate Nordic Secure DFU bootloader
- Implement flash configuration for safe updates
- Develop BLE-based OTA update protocol
- Create rollback mechanism for update failure recovery